Machine Learning
Gradient Descent

Step-by-Step Optimization of a Machine Learning Model

Key Idea

Gradient Descent is an optimization algorithm used to find parameter values that minimize the cost function.

$$\boxed{\text{Parameters} \rightarrow \text{Predictions} \rightarrow \text{Cost} \rightarrow \text{Gradient} \rightarrow \text{Parameter Update}}$$

The process is repeated until the model reaches a minimum or satisfies a chosen stopping condition.

1 Why Do We Need Gradient Descent?

A machine learning model contains parameters whose values must be learned from training data.

For Multiple Linear Regression:

$$\boxed{\hat{y} = w_0 + w_1 x_1 + w_2 x_2} \tag{1}$$

where:

- w_0 = bias
- w_1 = weight of feature x_1
- w_2 = weight of feature x_2
- $\hat{y}$ = predicted value

Initially, the best values of these parameters are unknown.

Therefore, the model needs a systematic way to find parameter values that produce a low cost.

$$\boxed{\text{Goal: Minimize the Cost Function}} \tag{2}$$

Gradient Descent provides one way to achieve this.

2 Example: Predicting Student Performance

Consider a simple machine learning problem:

Predict a student's exam score using two features.

The features are:

- x_1 = Hours Studied

- x_2 = Attendance

The target variable is:

$$y = \text{Exam Score}$$

For easy classroom calculation, the feature values are represented using small numerical values.

Student	**Hours** (x_1)	**Attendance** (x_2)	**Score** (y)
1	1	1	2
2	2	1	3
3	1	2	3
4	2	2	4
5	3	2	5
6	3	3	6

3 Step 1: Define the Model

For two input features, the Multiple Linear Regression model is:

$$\boxed{\hat{y} = w_0 + w_1 x_1 + w_2 x_2} \tag{3}$$

Here, the parameters to be learned are:

$$\boxed{w_0, \quad w_1, \quad w_2}$$

The model uses these parameters to calculate predictions.
For the i-th training example:

$$\boxed{\hat{y}_i = w_0 + w_1 x_{1i} + w_2 x_{2i}} \tag{4}$$

4 Step 2: Initialize the Parameters

Before training begins, the model needs initial values for its parameters.
For example:

$$w_0 = 0 \tag{5}$$

$$w_1 = 0 \tag{6}$$

$$w_2 = 0 \tag{7}$$

The learning rate is also selected:

$$\alpha = 0.1 \tag{8}$$

Key Idea

The initial parameter values do not have to be correct.
Gradient Descent will repeatedly modify them during training.

5 Step 3: Make Predictions

Using the current parameter values, the model calculates a prediction for every training example.

For observation i:

$$\boxed{\hat{y}_i = w_0 + w_1 x_{1i} + w_2 x_{2i}} \tag{9}$$

Therefore, every training example produces a prediction:

$$\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_n$$

These predictions are based on the current values of the parameters.

6 Step 4: Calculate the Error

For each training example, compare the actual value with the predicted value.

The error is:

$$\boxed{e_i = y_i - \hat{y}_i} \tag{10}$$

where:

- y_i = actual target value
- $\hat{y}_i$ = predicted value
- e_i = prediction error

7 Step 5: Calculate the Cost Function

The individual errors are combined to measure the overall performance of the model.

Using Mean Squared Error:

$$\boxed{J = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{11}$$

where:

$$n = \text{number of training examples}$$

The cost function gives us a single value representing the current error of the model.

Key Idea

$$\boxed{\text{Lower Cost} \Rightarrow \text{Better Fit to Training Data}}$$

During optimization, our goal is to reduce this value.

8 Step 6: Calculate the Gradient

The gradient tells us how the cost changes with respect to each parameter.

Since our model has three parameters,

$$w_0, \quad w_1, \quad w_2$$

we calculate a gradient for each parameter.

8.1 Gradient with Respect to Bias

$$\boxed{\frac{\partial J}{\partial w_0} = -\frac{2}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)} \tag{12}$$

8.2 Gradient with Respect to w_1

$$\boxed{\frac{\partial J}{\partial w_1} = -\frac{2}{n}\sum_{i=1}^{n} x_{1i}(y_i - \hat{y}_i)} \tag{13}$$

8.3 Gradient with Respect to w_2

$$\boxed{\frac{\partial J}{\partial w_2} = -\frac{2}{n}\sum_{i=1}^{n} x_{2i}(y_i - \hat{y}_i)} \tag{14}$$

Key Idea

Each parameter has its own gradient.
The gradient tells us how the cost changes when that particular parameter changes.

9 Understanding the Gradient

Suppose the gradient with respect to a parameter is:

$$\frac{\partial J}{\partial w_j} > 0$$

This means that increasing w_j tends to increase the cost.

Therefore, Gradient Descent moves w_j in the opposite direction.

If:

$$\frac{\partial J}{\partial w_j} < 0$$

the update moves w_j in the opposite direction again, which means the parameter increases.

Therefore:

Key Idea

Gradient Descent always attempts to move the parameters in the direction that reduces the cost.

10 Step 7: Update the Parameters

Once the gradients have been calculated, the parameters are updated.

The general Gradient Descent update rule is:

$$\boxed{w_j \leftarrow w_j - \alpha \frac{\partial J}{\partial w_j}} \tag{15}$$

For Multiple Linear Regression, each parameter is updated separately.

10.1 Update the Bias

$$\boxed{w_0 \leftarrow w_0 - \alpha \frac{\partial J}{\partial w_0}} \tag{16}$$

10.2 Update Weight w_1

$$\boxed{w_1 \leftarrow w_1 - \alpha \frac{\partial J}{\partial w_1}} \tag{17}$$

10.3 Update Weight w_2

$$\boxed{w_2 \leftarrow w_2 - \alpha \frac{\partial J}{\partial w_2}} \tag{18}$$

11 Why Do We Subtract the Gradient?

The gradient points in the direction in which the cost increases.

Our goal is to decrease the cost.

Therefore, we move in the opposite direction.

$$\boxed{\text{New Parameter} = \text{Old Parameter} - \text{Step in Gradient Direction}} \tag{19}$$

The step size is controlled by the learning rate:

$$\alpha$$

Therefore:

$$\boxed{\text{Parameter Update} = \alpha \times \text{Gradient}} \tag{20}$$

The negative sign makes the movement opposite to the gradient.

12 Step 8: Repeat the Process

After updating the parameters, the model uses the new values to make predictions again.

The complete cycle is:

1. Initialize parameters
↓
2. Make predictions
↓
3. Calculate errors
↓
4. Calculate cost
↓
5. Calculate gradients
↓
6. Update parameters
↓
7. Repeat

13 What Happens During Each Iteration?

One complete pass of the process can be represented as:

(w_0, w_1, w_2)
↓
Predictions
↓
Cost
↓
Gradients
↓
Updated (w_0, w_1, w_2)

After the update:

$$(w_0, w_1, w_2)_{\text{new}}$$

are used for the next iteration.

Therefore, the model continuously improves its parameter values.

14 Iteration

An **iteration** means one complete parameter update.

One iteration consists of:

Prediction → Cost → Gradient → Parameter Update

The process is repeated for multiple iterations.
For example:

$$\text{Iteration } 1 \rightarrow \text{Iteration } 2 \rightarrow \text{Iteration } 3 \rightarrow \cdots$$

Ideally, the cost decreases as training progresses.

15 Learning Rate

The learning rate α controls how large each parameter update is.

$$\boxed{\alpha = \text{Step Size}} \tag{21}$$

15.1 Learning Rate Too Small

If:

$$\alpha \rightarrow \text{very small}$$

the model takes very small steps.

Learning can become very slow.

15.2 Learning Rate Too Large

If:

$$\alpha \rightarrow \text{very large}$$

the model may take very large steps and overshoot the minimum.

The cost may fail to decrease properly.

15.3 Suitable Learning Rate

A suitable learning rate allows the model to move toward a minimum gradually.

16 Optimization View

Gradient Descent can be understood as an optimization problem.
We have a cost function:

$$J(w_0, w_1, w_2)$$

and we want to find:

$$\boxed{(w_0, w_1, w_2) \text{ that minimize } J} \tag{22}$$

Therefore:

$$\boxed{\min_{w_0, w_1, w_2} J(w_0, w_1, w_2)}$$

Gradient Descent searches for suitable parameter values by repeatedly moving in the direction that reduces the cost.

17 Optimization Intuition

Imagine the cost function as a landscape.

High Cost

↓

Move in the direction of decreasing cost

↓

Lower Cost

↓

Minimum

The model does not directly "know" the minimum.
Instead, it uses the gradient to decide how to move.

Key Idea

Gradient Descent is an iterative optimization process.
It gradually searches for parameter values that minimize the cost function.

18 Stopping the Optimization

Gradient Descent does not need to continue forever.
Training can stop when a suitable stopping condition is reached.
Common conditions include:

1. A fixed number of iterations is completed.
2. The change in cost becomes very small.
3. The gradient becomes sufficiently small.
4. The model reaches an acceptable level of convergence.

For example:

$$\left|J^{(t)} - J^{(t-1)}\right| < \epsilon \tag{23}$$

where ϵ is a small threshold.

19 Complete Gradient Descent Algorithm

1. **Choose the model.**

   For Multiple Linear Regression:

   $$\hat{y} = w_0 + w_1 x_1 + w_2 x_2$$

2. **Initialize parameters.**

   $$w_0, w_1, w_2$$

3. **Choose a learning rate.**

   $$\alpha$$

4. **Calculate predictions.**

   $$\hat{y}_i = w_0 + w_1 x_{1i} + w_2 x_{2i}$$

5. **Calculate the cost.**

   $$J = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

6. **Calculate the gradients.**

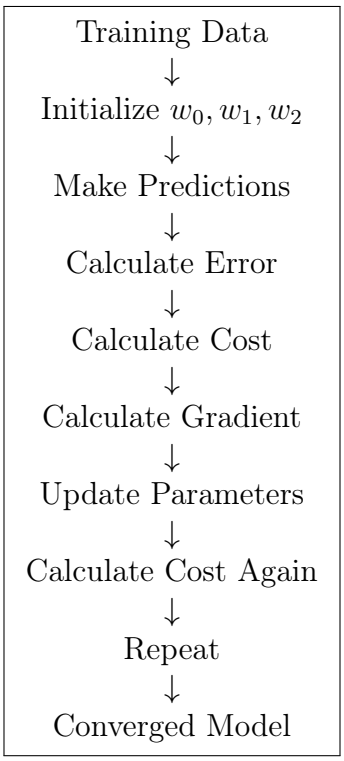
   $$\frac{\partial J}{\partial w_0}, \qquad \frac{\partial J}{\partial w_1}, \qquad \frac{\partial J}{\partial w_2}$$

7. **Update the parameters.**

   $$w_j \leftarrow w_j - \alpha \frac{\partial J}{\partial w_j}$$

8. **Repeat the process** until the chosen stopping condition is satisfied.

20 Gradient Descent: Complete Flow



21 Key Takeaways

1. Gradient Descent is an optimization algorithm.
2. Its objective is to minimize the cost function.
3. The model starts with initial parameter values.
4. Predictions are calculated using the current parameters.
5. The cost measures the overall prediction error.
6. The gradient tells us how the cost changes with respect to each parameter.
7. Parameters are updated in the opposite direction of the gradient.
8. The learning rate controls the size of the update.
9. The process is repeated over multiple iterations.
10. Training stops when an appropriate stopping condition is reached.

22 Quick Revision

Term	Meaning
Parameter	A value learned by the model
Weight	Parameter controlling the contribution of a feature
Bias	Intercept or baseline parameter
Cost Function	Measures overall error of the model
Gradient	Shows how cost changes with respect to a parameter
Learning Rate	Controls the size of each update
Iteration	One complete parameter update
Gradient Descent	Optimization algorithm used to minimize cost
Convergence	Condition where further updates produce very little improvement

One-Line Summary

Gradient Descent repeatedly updates parameters to minimize the cost fun